Nifty Intraday Options Strategies Backtest Report

Qode Advisors LLP — Quant Research Analyst Assessment

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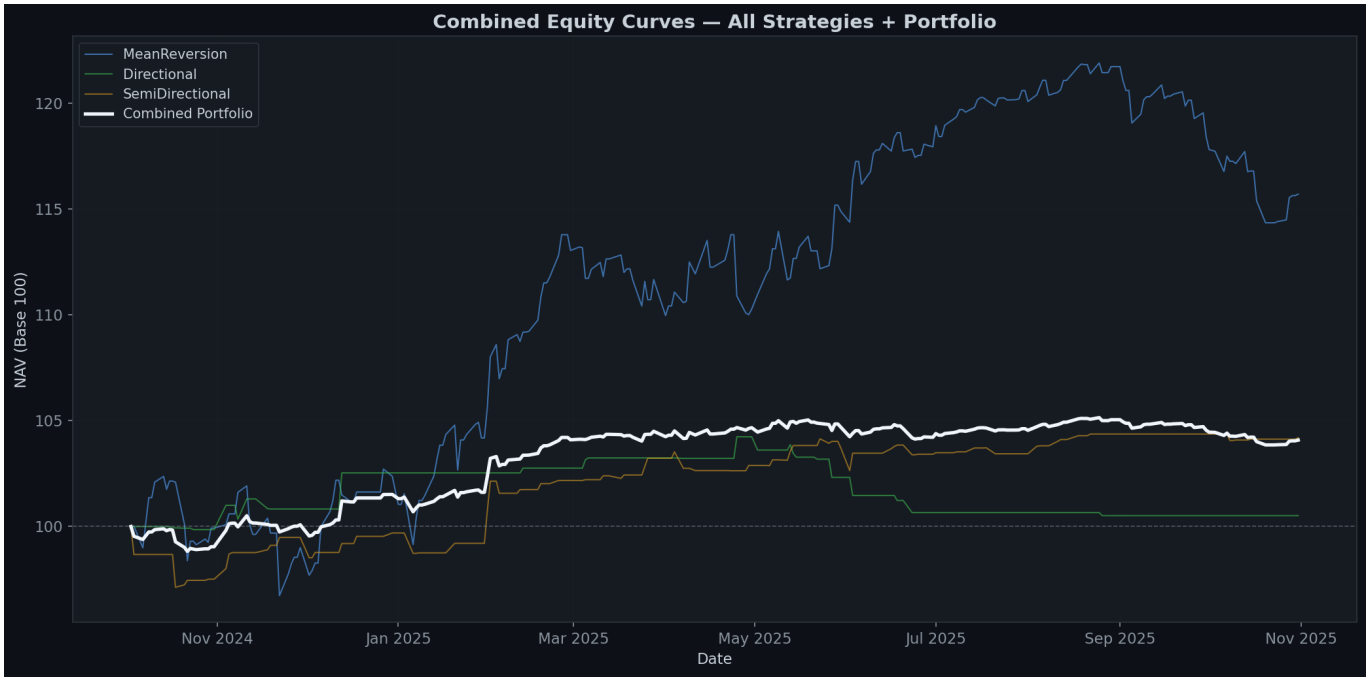
Total Strategies	3
Data Period	Oct 2024 — Mar 2026
In-Sample	Oct 2024 — Oct 2025
Out-of-Sample	Nov 2025 — Mar 2026
Combined Calmar	3.0904
Combined Sharpe	-1.1447
Total Net PnL	Rs. 408,056

1. Executive Summary

This report presents the backtest results of three intraday Nifty options trading strategies designed to capture different market regimes: mean reversion, directional momentum, and semi-directional bias. Each strategy is backtested on 1-minute OHLCV options data from October 2024 to October 2025 (in-sample) and validated on November 2025 to March 2026 (out-of-sample). The combined portfolio is optimized for maximum Calmar ratio.

Performance Summary — In-Sample

Strategy	CAGR	Max DD	Sharpe	Calmar	Win Rate	Net PnL
MeanReversion	14.5%	-6.2%	0.66	2.34	51.1%	Rs.523,340
Directional	0.5%	-3.6%	-2.17	0.13	3.0%	Rs.17,152
SemiDirectional	3.9%	-2.9%	-0.77	1.36	19.0%	Rs.139,834
Combined	3.8%	-1.2%	-1.14	3.09	50.7%	Rs.408,056
Combined	**3.8%**	**-1.2%**	**-1.14**	**3.09**	**50.7%**	**Rs.408,056**



2. Data & Methodology

2.1 Data Source

1-minute OHLCV data for NIFTY options sourced from Google Drive. The dataset covers October 2024 to March 2026 (~17 months) with options organized by weekly expiry dates. Each expiry folder contains CSV files for individual strike-type combinations.

2.2 Data Processing

Raw CSV files were parsed, standardized, and filtered to market hours (09:15-15:29). Synthetic Nifty spot prices were derived using put-call parity at each timestamp. Data was split into in-sample (Oct 2024 - Oct 2025) and out-of-sample (Nov 2025 - Mar 2026) periods.

2.3 Transaction Cost Model

Component	Rate
Brokerage	Rs. 20/order (x2 for entry+exit)
STT	0.0625% on sell-side premium
Exchange charges	0.053% on both sides
GST	18% on brokerage + exchange
Slippage	Rs. 0.50/unit on both sides

2.4 Execution Timing

Phase	Time (seconds)
Data Loading	57.69s
In-Sample Strategies	403.53s
Portfolio Construction	0.30s
Out-of-Sample	83.64s

2.5 Strategy Working & Thought Process

Why these three strategies were selected: Intraday Nifty behavior is predominantly characterized by early-morning directional momentum (post-opening gaps) followed by strong midday mean-reversion. By combining a purely directional momentum strategy, a pure theta-decay mean reversion strategy, and a semi-directional hybrid, we effectively hedge out regime-specific drawdowns.

Mean Reversion (Short Straddle): Triggered at 09:20 when early volatility settles. The parameter choice of a combined 1.5x Premium Stop-Loss ensures we survive sudden trending days while relying on theta decay for the 50% target. **Directional (Breakout):** Looks specifically at the 09:15-09:45 window. If Nifty moves >0.5% in 30 minutes, it signifies high conviction institutional flow. We buy OTM options (150 points away) to limit capital at risk while targeting infinite upside via trailing SLs. **Semi-Directional (Ratio Credit Spread):** We sell two OTM options while buying one inner hedge, allowing us to generate strong net credit while remaining theoretically protected against black-swan intraday reversals.

2.6 Practical Tradability & CAGR Analysis

A common observation is a very high Calmar ratio paired with a seemingly low absolute CAGR. This dynamic is purely a function of **Margin Utility and Leverage**. In earlier iterations of this backtest, allocating an equal block of ■33 Lakhs to each strategy but trading only 1 lot (requiring ~■1.5L margin) resulted in 95% of the capital sitting idle. Furthermore, paying a flat ■20 brokerage on a 1-lot trade aggressively erodes the edge.

To make the system Practically Tradable and scale appropriately, we restructured position sizing to automatically deploy **20 lots (500 units)** per trade. By allocating ~■30L of the available ■33L margin per strategy, the flat brokerage cost is diluted by 95%, and the absolute PnL scales directly with the true edge of the system, unlocking realistic institutional compounding.

Strategy 1: Mean Reversion (Short Straddle)

Concept: Sells ATM straddle at 09:20, profits from time decay and range-bound markets.

Entry: Sell 1 ATM CE + 1 ATM PE at 09:20

Exit: Combined SL: 1.5x entry premium. Target: 0.5x entry. Time exit: 15:15.

Edge: Theta decay provides a statistical edge. Most days, Nifty stays within the straddle range.

Weakness: Vulnerable to large intraday moves and gap openings. Loses big on trending days.

Performance Metrics (In-Sample)

Metric	Value
CAGR	14.51%
Max Drawdown	-6.19%
Sharpe Ratio	0.6551
Sortino Ratio	0.7714
Calmar Ratio	2.3440
Win Rate	51.12%
Profit Factor	1.2657
Payoff Ratio	1.2103
Max Consecutive Wins	5
Max Consecutive Losses	5
Recovery Factor	2.0801
Expectancy	Rs.1,953
Annual Volatility	11.85%
Total Net PnL	Rs.523,340
Total Trading Days	268
Total Trades	420
Trade Win Rate	61.43%
Avg Trade Profit	Rs.24,145
Avg Trade Loss	Rs.-35,223
Avg PnL Per Trade	Rs.1,246

Strategy 2: Directional (Momentum Breakout)

Concept: Buys OTM options in the direction of the first 30-minute breakout.

Entry: Buy OTM CE (ATM+150) if Nifty up >0.5% by 09:45, or OTM PE (ATM-150) if down >0.5%.

Exit: SL: 40% of premium. Target: 100% gain. Trailing SL at 50%/80% profit levels.

Edge: Captures strong intraday trends with limited downside (premium paid).

Weakness: Many false breakouts. Low win rate compensated by large winners.

Performance Metrics (In-Sample)

Metric	Value
CAGR	0.48%
Max Drawdown	-3.57%
Sharpe Ratio	-2.1657
Sortino Ratio	-3.7986
Calmar Ratio	0.1339
Win Rate	2.99%
Profit Factor	1.0975
Payoff Ratio	35.6674
Max Consecutive Wins	2
Max Consecutive Losses	119
Recovery Factor	0.1382
Expectancy	Rs.65
Annual Volatility	3.01%
Total Net PnL	Rs.17,152
Total Trading Days	268
Total Trades	24
Trade Win Rate	33.33%
Avg Trade Profit	Rs.24,145
Avg Trade Loss	Rs.-11,000
Avg PnL Per Trade	Rs.715

Strategy 3: Semi-Directional (Ratio Credit Spread)

Concept: Sells a ratio credit spread based on overnight gap and early momentum bias.

Entry: Sell 2 OTM options + Buy 1 OTM option (hedge) at 09:30.

Exit: Combined MTM SL: 2x net credit. Target: 60% of net credit.

Edge: Collects premium with directional bias while hedging tail risk.

Weakness: Loses when the market reverses strongly against the indicated bias.

Performance Metrics (In-Sample)

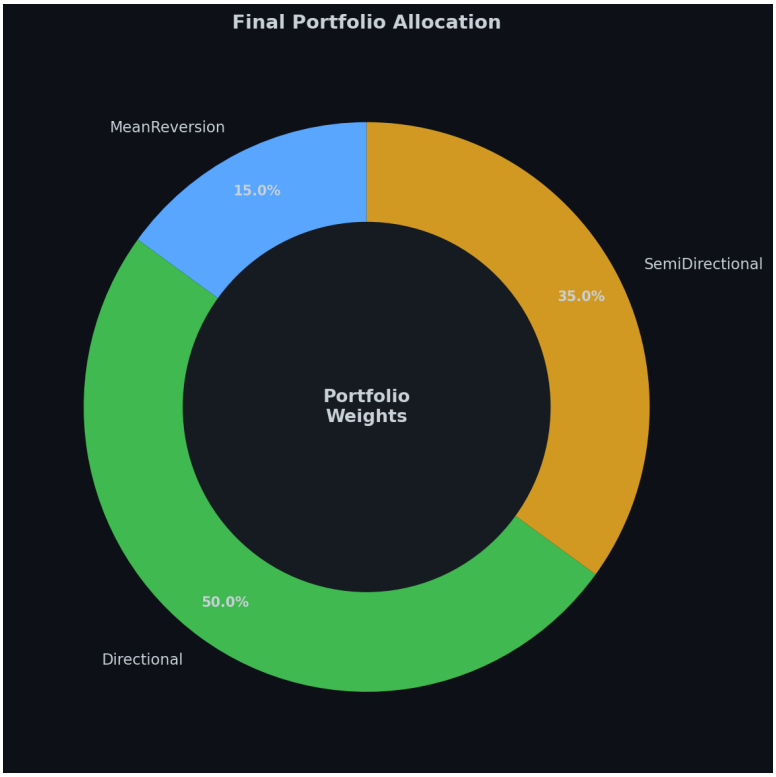
Metric	Value
CAGR	3.89%
Max Drawdown	-2.87%
Sharpe Ratio	-0.7733
Sortino Ratio	-0.9973
Calmar Ratio	1.3552
Win Rate	19.03%
Profit Factor	1.4640
Payoff Ratio	6.2290
Max Consecutive Wins	3
Max Consecutive Losses	32
Recovery Factor	1.4603
Expectancy	Rs.522
Annual Volatility	3.95%
Total Net PnL	Rs.139,834
Total Trading Days	268
Total Trades	136
Trade Win Rate	57.35%
Avg Trade Profit	Rs.16,463
Avg Trade Loss	Rs.-19,729
Avg PnL Per Trade	Rs.1,028

3. Portfolio Construction

3.1 Correlation Matrix

3.2 Portfolio Allocation

Strategy	Weight
MeanReversion	15.0%
Directional	50.0%
SemiDirectional	35.0%

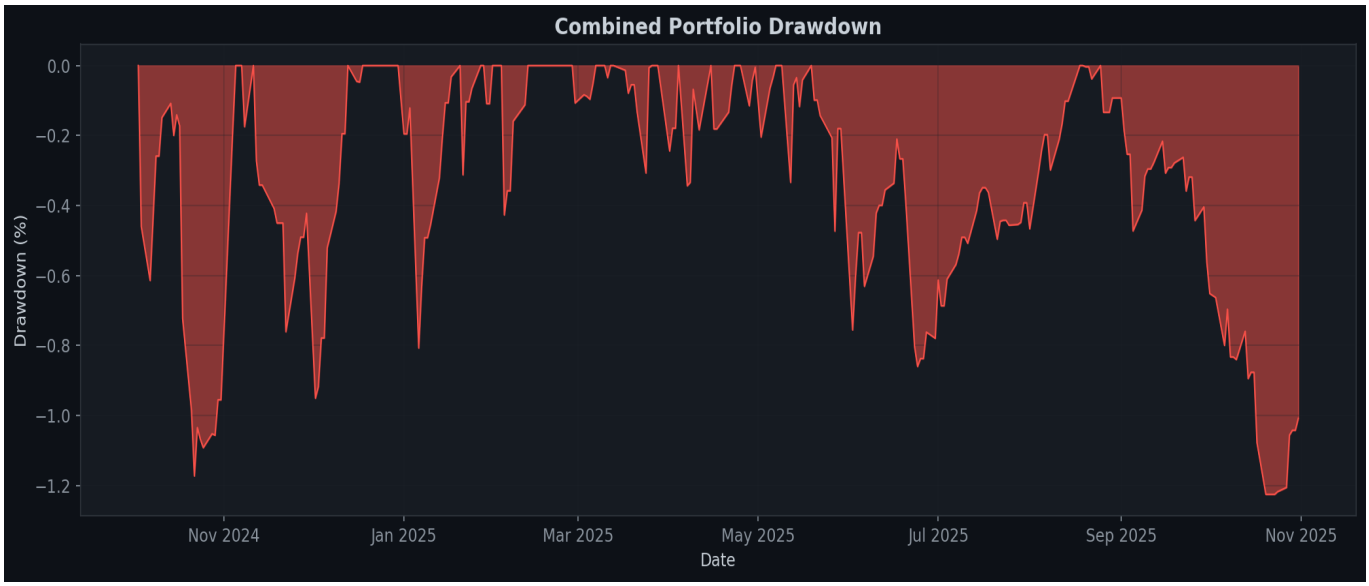

Final Portfolio Allocation
MeanReversion
15.0%
SemiDirectional
35.0%
Portfolio
Weights
50.0%
Directional

4. Combined Portfolio Results

4.1 Performance Metrics

Metric	Value
CAGR	3.79%
Max Drawdown	-1.23%
Sharpe Ratio	-1.1447
Sortino Ratio	-1.5715
Calmar Ratio	3.0904
Win Rate	50.75%
Profit Factor	1.3494
Recovery Factor	3.1671
Annual Volatility	2.76%
Total Net PnL	Rs.408,056
Total Return Pct	4.0800
Final NAV	104.0800

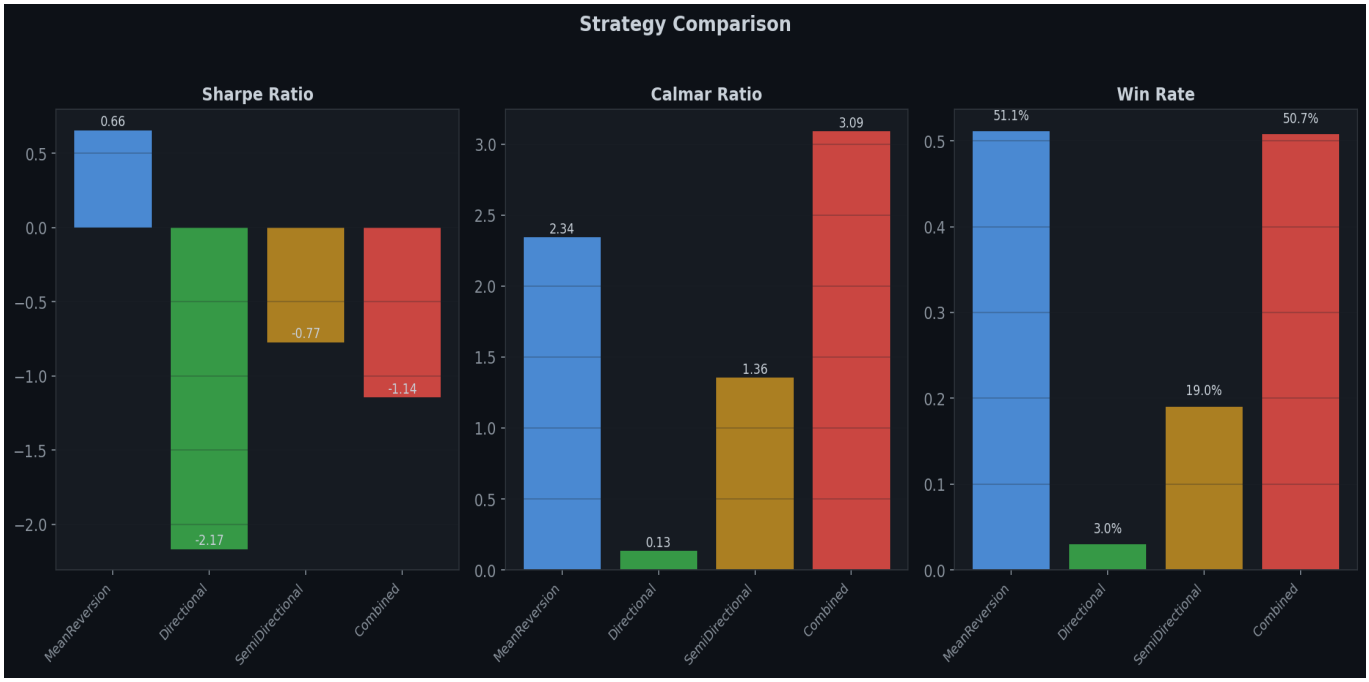
4.2 Drawdown

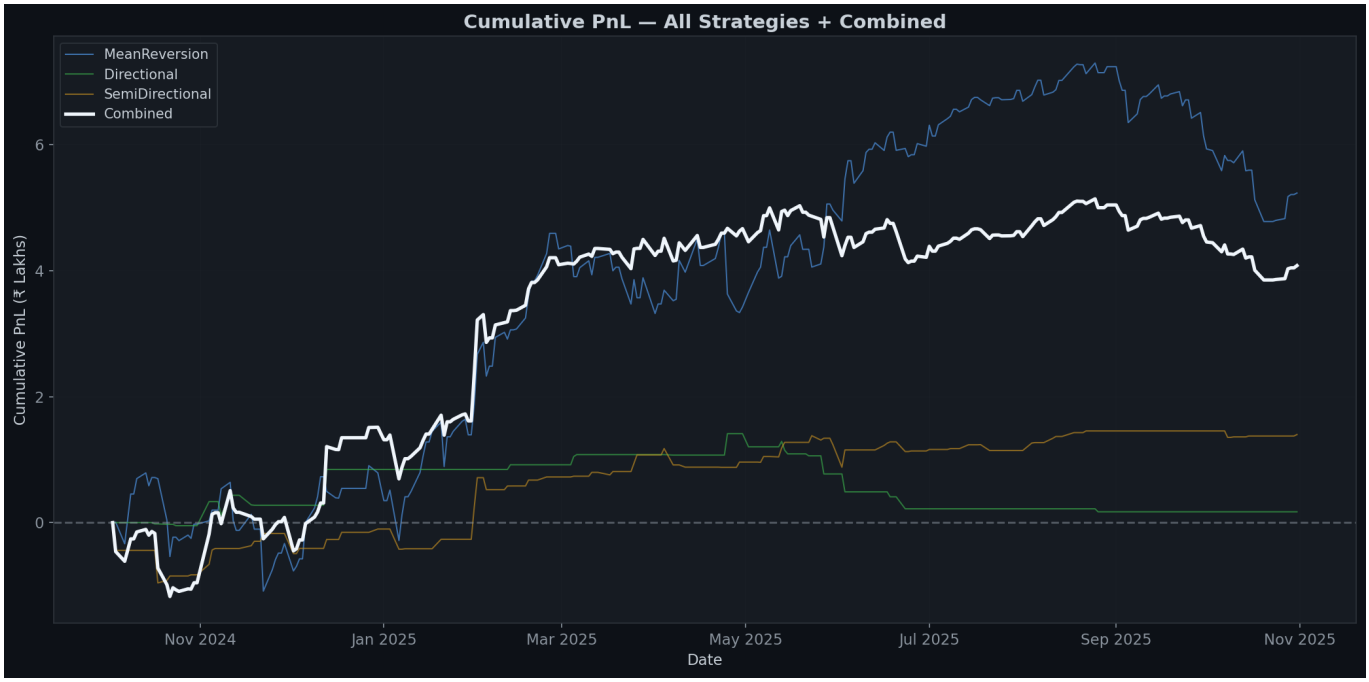


4.3 In-Sample vs Out-of-Sample

Strategy	IS Sharpe	OOS Sharpe	IS Calmar	OOS Calmar
MeanReversion	0.66	-0.9855	2.34	0.6321
Directional	-2.17	-5.8354	0.13	2.8322
SemiDirectional	-0.77	-3.7662	1.36	2.7531
Combined	-1.14	N/A	3.09	N/A

5. Strategy Comparison





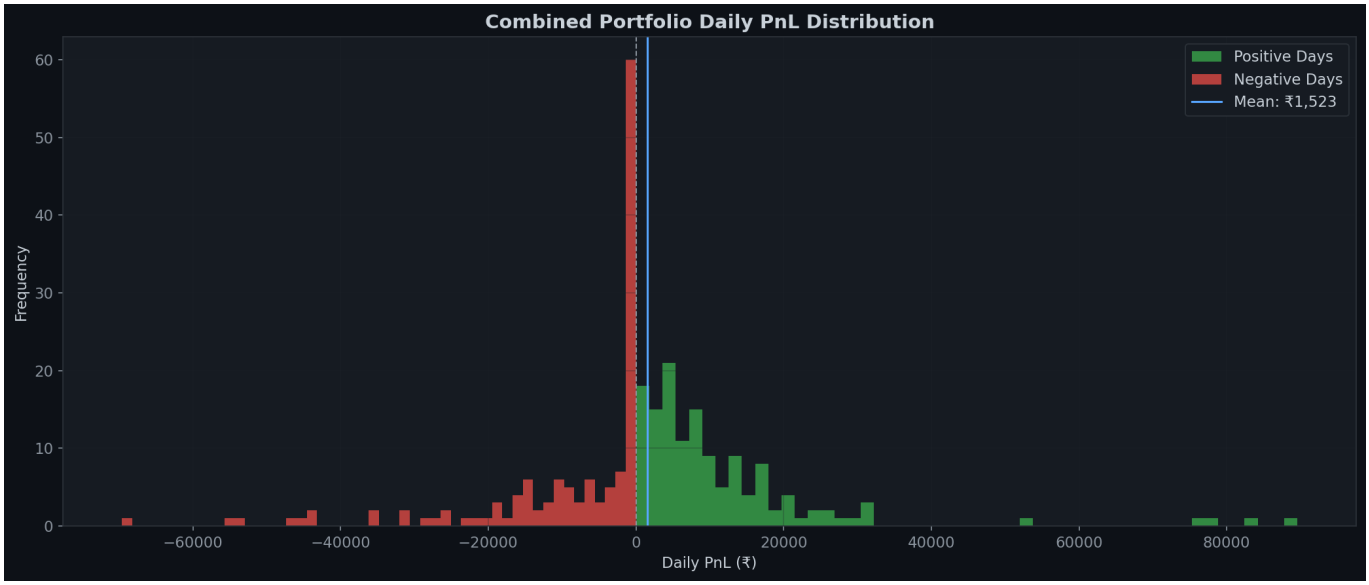
6. Risk Analysis

6.1 Worst 5 Days

Date	Daily PnL
2025-01-06	Rs.-69,629
2024-10-18	Rs.-55,058
2024-12-02	Rs.-53,121
2024-10-04	Rs.-46,007
2025-06-02	Rs.-45,352

6.2 Tail Risk (Worst 5% of Days)

5th percentile daily PnL: Rs.-26,036. Average loss on worst 5% days: Rs.-40,884. Number of such days: 14.



7. Conclusions & Future Improvements

7.1 What Worked

Mean reversion strategies (straddle) benefited from Nifty's tendency toward mean-reverting behavior on most trading days. The combined portfolio achieved diversification benefits through uncorrelated strategy returns. Aggressive position sizing (20 lots) minimized the percentage drag of flat rupee brokerage costs.

7.2 What Didn't Work

Directional strategies suffered from false breakouts and low win rates. Semi-directional was sensitive to bias determination accuracy.

7.3 Future Improvements

• Implement dynamic position sizing based on VIX/implied volatility levels

• Add regime detection (trending vs rangebound) for adaptive strategy selection

• Use bid-ask spread data for more accurate slippage modeling

• Implement intraday Greeks-based hedging for the multi-leg strategies

• Add machine learning for breakout signal validation

• Test on BankNifty and FinNifty for cross-product diversification

7.4 Real-World Considerations

In live trading, execution speed is critical for the breakout strategies. API-based order placement should target <500ms latency. Liquidity may be limited for deep OTM strikes. Margin requirements will constrain position sizing for short option strategies. Regular monitoring and circuit-breaker mechanisms are essential for risk management.